QUESTION-TO-SLIDE MAPPING

SECTION A - LECTURE 09

Shading Device

Course	BECM 3115 - Climate and Architectural Design
Lecture deck	9.0 - Shading Device (12 PDF pages/slides)
Question bank	Regular term-final examinations, 2016-2024 (8 supplied papers; 2019 not present)
Method	Topic-level mapping to the 12-slide Lecture 09 deck. The deck is credited for the purpose of shading, direct/reflected/diffuse solar paths, external-versus-internal shading logic, horizontal/vertical/combined devices, traditional examples and effective-shading concepts. The 2023-2024 solar-shading numericals are marked Partial because the deck does not provide the solar-position/projection calculation chain.

RESULT

Lecture 09 is the dedicated shading answer source. It directly supports the 2016 effective-shading and shading-type questions and the 2021-2022 indoor-comfort-by-shading questions. It also supports the 2023-2024 seasonal-overhang questions conceptually, while the 15-17 mark numerical projection problems require additional solar-geometry method not present in these slides.

Prepared as part of the cumulative lecture-by-lecture question bank analysis.

1. Executive Summary

Lecture 09 explains why and how shading devices control solar radiation, wind, rainwater and visibility. It distinguishes direct, reflected and diffuse radiation through openings, shows why external shading is thermally superior to internal shading, classifies horizontal, vertical and combined devices, provides traditional shading examples, and closes with an effective-shading comparison. These are highly sketch-oriented exam topics.

12	9	4	2	3	112
Slides reviewed	Question links	Exact matches	Direct matches	Partial matches	Mapped marks

Lecture structure and exam value

Slides	Lecture block	Historical signals	Exam value
2-5	Purpose, solar paths, external vs internal shading	2016 Q7(c); 2021/2022 Q8(a)	Core concept block: stop heat before it enters, understand direct/reflected/diffuse paths.
6-8	Horizontal / vertical / combined types	2016 Q8(a); 2021/2022 Q8(a)	Exact classification and orientation-sensitive sketches.
9-10	Modern/traditional examples	Supports theory answers	Use examples to enrich answers; not a separate historical question family.
11	Effective shading alternatives + reduction figures	2016 Q7(c); 2021/2022 Q8(a)	Useful comparison slide; treat percentages as slide-specific indicative values.
12	Closing	No direct question	No study time needed.

Priority note Slides 2-8 and 11 are the scoring core. In the exam, always draw the sun direction, the window/opening and the blocked ray; a device shape without ray logic is a weak answer.

2. Detailed Year-wise Question Mapping

Original paper references are preserved. Match labels mean: Exact = the slide block substantially answers the task; Direct = the requested concept is clearly taught but some expansion/sketch/application may still be needed; Partial = the deck only supports part of the requested answer.

Year	Original	Marks	QB p.	Question / task	Match	Lecture slides	Coverage assessment
2016	Q7(c)	10	1	What is effective shading? Explain with necessary sketches.	Exact	2-5, 11	The deck defines the purpose, shows external/internal effectiveness and provides an effective-shading comparison with sketches.
2016	Q8(a)	15	1	Specify various types of shading devices with necessary sketches.	Exact	6-10	Horizontal, vertical and combined devices are explicitly classified and illustrated.
2021	Q8(a)	12	7	How can indoor climatic comfort be ensured by designing shading devices?	Exact	2-11	The deck covers solar, wind, rain and visibility control plus device types and effectiveness.
2022	Q8(a)	10	9	How can climatic comfort be ensured in a building by designing shading devices?	Exact	2-11	Direct repeat of the lecture purpose and device-selection logic.
2023	Q6(a)	6	11	Show how an extended overhead/lofty porch blocks summer glare but admits winter sun.	Direct	3-5, 7	Horizontal external shading concept is directly supported; the exact seasonal-ray sketch is not separately taught.

2. Detailed Year-wise Question Mapping - Continued

Year	Original	Marks	QB p.	Question / task	Match	Lecture slides	Coverage assessment
2023	Q7(a)	15	11	Solve a solar-shading problem for Khulna and determine effective horizontal/vertical projection and incidence angle.	Partial	6-8, 11	The deck identifies horizontal/vertical devices and effective shading, but it does not contain solar-position or projection equations.
2024	Q5(c)	5	14	Explain how an extended lofty porch/overhead shade blocks summer glare and admits winter sun.	Direct	3-5, 7	Conceptual horizontal-overhang answer is supported; seasonal geometry should be completed from the taught sun-angle method.
2024	Q7(b)	17	15	Solve a solar-shading projection problem and determine incidence angle.	Partial	6-8, 11	Device concepts are present, but the analytical solar-geometry chain is absent from Lecture 09.
2024	Q8(b)	10	15	Explain building-envelope response in microclimate design.	Partial	2-5	Shading is one important envelope response, but the full component answer belongs to Lecture 08.

3. Repeated Question Patterns and Quick Recognition

These patterns compress historical wording into the question families that matter for revision.

P	Normalized repeated question	Years seen	Slides	Why it matters
S	Effective shading / indoor climatic comfort by shading	2016, 2021, 2022	2-5, 11	Repeated theory family; easy to score with external-vs-internal sketch.
S	Types of shading devices with sketches	2016 + later related	6-10	Memorize horizontal, vertical, combined and orientation logic.
A	Seasonal overhang / lofty porch	2023, 2024	3-5, 7	Recent short question; combine lecture concept with high-summer/low-winter ray sketch.
A	Solar shading projection numerical	2023, 2024	6-8, 11 partial	Recent 15-17 mark trend, but calculations are outside the deck.

Quick recognition rules

If the question says...	Go to slides	Recognition rule
“effective shading”	2-5, 11	Start with external interception before glazing; compare internal shading only after that.
“types of shading devices”	6-8	Horizontal = high-angle control; vertical = low-angle side sun; combined = both.
“summer glare / winter sun / lofty porch”	3-5, 7	Draw a horizontal overhang with high summer ray blocked and low winter ray admitted.
“projection depth / incidence angle”	6-8, 11 + external method	Recognize it as numerical solar geometry; do not invent equations from these slides.
“rain / privacy / wind control”	2	Mention these as additional functions after solar control.

4. Slide-to-Question Reverse Index

Use this page when studying directly from the lecture deck: open the slide block, then practice the linked question family immediately.

Slides	What the slides teach	Past-question links	Study action
2	Shading controls solar, wind, rain, visibility	2021 Q8(a); 2022 Q8(a)	Use as four-function introduction.
3-4	Direct/reflected/diffuse solar paths	2016 Q7(c); 2021/2022 Q8(a)	Draw source-path-opening diagram.
5	External vs internal shading effectiveness	2016 Q7(c); 2021/2022 Q8(a)	Must-know thermal argument: stop radiation before glazing.
6-8	Horizontal, vertical, combined types	2016 Q8(a); 2023/2024 seasonal/numerical support	Practice three simple labelled device sketches.
9-10	Applied/traditional examples	Theory enrichment	Use one or two examples only.
11	Effective-shading alternatives	2016 Q7(c); 2021/2022 Q8(a)	Compare fins/louvres/awning; percentages are slide-specific, not universal constants.
12	Closing	None	Skip.

5. Coverage Gaps and Diagnostic Exclusions

Lecture 09 is a complete theory source for shading devices, but it is not a solar-geometry calculation lecture. That distinction is essential for the recent numerical questions.

Gap	Affected questions	Action
Solar altitude/azimuth, HSA/profile angle, projection sizing and incidence-angle equations	2023 Q7(a); 2024 Q7(b)	Use the dedicated solar-shading analytical method / class method. Do not derive these from Lecture 09 alone.
Exact seasonal sun-angle construction for lofty porch	2023 Q6(a); 2024 Q5(c)	Lecture 09 supports horizontal shading concept; add the taught high-summer/low-winter sun sketch.
Detailed building-envelope mechanisms beyond shading	2024 Q8(b) and repeated envelope family	Use Lecture 08.
Cosine Law derivation/application	2024 Q6(c) and earlier cosine-law questions	Use Lecture 05; it is not part of this deck.

Mapping discipline	The effective-shading percentages shown on Slide 11 are treated as lecture examples, not universal performance guarantees. Numerical solar-geometry questions remain Partial because the required equations and solar-position procedure are absent from the deck.

6. Recommended Study Plan

Study this lecture in three passes: purpose/solar paths, device classification, then effectiveness/examples. Finish by practicing the two recent seasonal/numerical question types with the appropriate companion method.

Step	Study block	Slides	Practice immediately
1	Effective shading + external vs internal	2-5, 11	2016 Q7(c); 2021 Q8(a); 2022 Q8(a)
2	Horizontal / vertical / combined types	6-8	2016 Q8(a)
3	Seasonal overhang concept	3-5, 7	2023 Q6(a); 2024 Q5(c)
4	Traditional/applied examples	9-10	Add 1-2 examples to theory answers
5	Numerical handoff	6-8, 11 + solar-geometry method	2023 Q7(a); 2024 Q7(b)

Exam-ready minimum from this lecture

Priority	What to reproduce	Exam technique
A	External vs internal shading sketch	Show solar ray blocked outside versus heat admitted through glazing.
A	Three basic device sketches	Horizontal, vertical, combined with sun direction.
A	Direct / reflected / diffuse paths	Simple source diagram with labels.
B	Seasonal overhang sketch	High summer ray blocked; low winter ray admitted.

Bottom line	Lecture 09 is a compact, high-return theory lecture. Master the ray logic and three device types first; then keep the 2023-2024 numerical questions clearly separated as a companion solar-geometry skill.